THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(c) $\frac{1}{4}$	Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$; $P = \frac{9}{36} = \frac{1}{4}$
2	(b) $\frac{7}{8}$	Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$; $P = \frac{7}{8}$
3	(b) $\frac{7}{15}$	Favourable numbers: 3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30; number $= 14$; $P = \frac{14}{30} = \frac{7}{15}$
4	(b) 6	$\frac{x}{3+x} = 2 \times \frac{3}{3+x}$ gives $x = 2 \times 3$; $x = 6$
5	(d) 2	Cross-multiplying: $2(3 + x) = 1(8 + x)$; $x = 2$
6	(d) $\frac{1}{4}$	Favourable outcomes: (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6); number $= 9$; $P = \frac{9}{36} = \frac{1}{4}$
7	(d) $\frac{5}{11}$	Favourable letters: E, U, A, I, O; number $= 5$; $P = \frac{5}{11}$
8	(b) $\frac{5}{12}$	Favourable outcomes: 15 pairs; number $= 15$; $P = \frac{15}{36} = \frac{5}{12}$
9	(d) $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, HTT; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
10	(b) $\frac{27}{40}$	Favourable numbers: 27 outcomes; number $= 27$; $P = \frac{27}{40}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,2), (1,5), (2,1), (2,4), (3,3), (3,6), (4,2), (4,5), (5,1), (5,4), (6,3), (6,6); number $= 12$ (1); $P = \frac{12}{36} = \frac{1}{3}$ (1).	1+1
12	Total possible outcomes $= 36$; Favourable numbers: 6, 12, 18, 24, 30, 36; number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1
13	Total balls $= 6 + 4 + 5 = 15$; Favourable balls $= 10$ (1); $P = \frac{10}{15} = \frac{2}{3}$ (1).	1+1
14	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,2), (2,1), (2,4), (3,6), (4,2), (6,3); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{blue}) = \frac{x}{7+x}$ and $P(\text{red}) = \frac{7}{7+x}$ (1); $\frac{x}{7+x} = 2 \times \frac{7}{7+x}$ gives $x = 2 \times 7$ (1); $x = 14$ (1).	1+1+1
16	$\frac{3+x}{9+x} = \frac{1}{2}$ (1); Cross-multiplying: $2(3 + x) = 1(9 + x)$ (1); $x = 3$ (1).	1+1+1
17	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$ (1); $P = \frac{5}{36}$ (1).	1+1+1
	OR: $P(\text{blue}) = \frac{x}{8+x}$ and $P(\text{red}) = \frac{8}{8+x}$ (1); $\frac{x}{8+x} = 3 \times \frac{8}{8+x}$ gives $x = 3 \times 8$ (1); $x = 24$ (1).	1+1+1
18	$\frac{5+x}{10+x} = \frac{2}{3}$ (1); Cross-multiplying: $3(5 + x) = 2(10 + x)$ (1); $x = 5$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 4; 15; 12 (2); Probabilities: $\frac{1}{9}$, $\frac{5}{12}$, $\frac{1}{3}$ (1).	2+2+1
	OR: Total outcomes $= 40$ (2); Favourable counts: 13, 12, 10 (2); Probabilities: $\frac{13}{40}$, $\frac{3}{10}$, $\frac{1}{4}$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 6; 14; 7 (2); Probabilities: $\frac{1}{6}$, $\frac{7}{18}$, $\frac{7}{36}$ (1).	2+2+1
	OR: Total outcomes $= 50$ (2); Favourable counts: 7, 10, 1 (2); Probabilities: $\frac{7}{50}$, $\frac{1}{5}$, $\frac{1}{50}$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.

- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	$\frac{x}{4+x} = 3 \times \frac{4}{4+x}$ gives $x = 3 \times 4$; $x = 12$
2	**(a)** 6	Cross-multiplying: $8(4 + x) = 5(10 + x)$; $x = 6$
3	**(b)** $\frac{11}{36}$	Favourable outcomes: (1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6); number $= 11$; $P = \frac{11}{36}$
4	**(b)** $\frac{4}{11}$	Favourable letters: O, A, I, I; number $= 4$; $P = \frac{4}{11}$
5	**(c)** $\frac{1}{36}$	Favourable outcomes: (6,6); number $= 1$; $P = \frac{1}{36}$
6	**(c)** $\frac{1}{4}$	Favourable outcomes: HHH, HTH; number $= 2$; $P = \frac{2}{8} = \frac{1}{4}$
7	**(b)** $\frac{13}{50}$	Favourable numbers: 2, 6, 10, 14, 18, 22, 26, 30, 34, 38, 42, 46, 50; number $= 13$; $P = \frac{13}{50}$
8	**(c)** 10	$\frac{x}{5+x} = 2 \times \frac{5}{5+x}$ gives $x = 2 \times 5$; $x = 10$
9	**(b)** 7	Cross-multiplying: $7(5 + x) = 4(14 + x)$; $x = 7$
10	**(b)** $\frac{1}{9}$	Favourable outcomes: (1,2), (2,1), (5,6), (6,5); number $= 4$; $P = \frac{4}{36} = \frac{1}{9}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total possible outcomes $= 42$; Favourable numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41; number $= 13$ (1); $P = \frac{13}{42}$ (1).	1+1
12	Total balls $= 5 + 9 + 6 = 20$; Favourable balls $= 14$ (1); $P = \frac{14}{20} = \frac{7}{10}$ (1).	1+1
13	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: 14 pairs; number $= 14$ (1); $P = \frac{14}{36} = \frac{7}{18}$ (1).	1+1
14	Total possible outcomes $= 48$; Favourable numbers: 1, 4, 9, 16, 25, 36; number $= 6$ (1); $P = \frac{6}{48} = \frac{1}{8}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{blue}) = \frac{x}{9+x}$ and $P(\text{red}) = \frac{9}{9+x}$ (1); $\frac{x}{9+x} = 2 \times \frac{9}{9+x}$ gives $x = 2 \times 9$ (1); $x = 18$ (1).	1+1+1
16	$\frac{4+x}{16+x} = \frac{2}{5}$ (1); Cross-multiplying: $5(4 + x) = 2(16 + x)$ (1); $x = 4$ (1).	1+1+1
17	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1+1
	OR: $P(\text{blue}) = \frac{x}{10+x}$ and $P(\text{red}) = \frac{10}{10+x}$ (1); $\frac{x}{10+x} = 4 \times \frac{10}{10+x}$ gives $x = 4 \times 10$ (1); $x = 40$ (1).	1+1+1
18	$\frac{7+x}{10+x} = \frac{17}{20}$ (1); Cross-multiplying: $20(7 + x) = 17(10 + x)$ (1); $x = 10$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (2,6), (3,4), (4,3), (6,2); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 5; 15; 11 (2); Probabilities: $\frac{5}{36}$, $\frac{5}{12}$, $\frac{11}{36}$ (1).	2+2+1
	OR: Total outcomes $= 36$ (2); Favourable counts: 6, 17, 24 (2); Probabilities: $\frac{1}{6}$, $\frac{17}{36}$, $\frac{2}{3}$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 6; 4; 25 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{9}$, $\frac{25}{36}$ (1).	2+2+1
	OR: Total outcomes $= 60$ (2); Favourable counts: 15, 52, 48 (2); Probabilities: $\frac{1}{4}$, $\frac{13}{15}$, $\frac{4}{5}$ (1).	2+2+1

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- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{2}{11}$	Favourable letters: C, C; number $= 2$; $P = \frac{2}{11}$
2	**(a)** $\frac{1}{9}$	Favourable outcomes: (1,4), (2,3), (3,2), (4,1); number $= 4$; $P = \frac{4}{36} = \frac{1}{9}$
3	**(c)** $\frac{7}{8}$	Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$; $P = \frac{7}{8}$
4	**(c)** $\frac{13}{15}$	Favourable numbers: 39 outcomes; number $= 39$; $P = \frac{39}{45} = \frac{13}{15}$
5	**(b)** 24	$\frac{x}{6+x} = 4 \times \frac{6}{6+x}$ gives $x = 4 \times 6$; $x = 24$
6	**(c)** 4	Cross-multiplying: $2(2 + x) = 1(8 + x)$; $x = 4$
7	**(b)** $\frac{25}{36}$	Favourable outcomes: 25 pairs; number $= 25$; $P = \frac{25}{36}$
8	**(a)** $\frac{5}{8}$	Favourable letters: T, R, N, G, L; number $= 5$; $P = \frac{5}{8}$
9	**(b)** $\frac{1}{9}$	Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$; $P = \frac{4}{36} = \frac{1}{9}$
10	**(c)** $\frac{1}{2}$	Favourable outcomes: HHH, HHT, TTH, TTT; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total balls $= 8 + 2 + 10 = 20$; Favourable balls $= 18$ (1); $P = \frac{18}{20} = \frac{9}{10}$ (1).	1+1
12	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,4), (2,4), (3,4), (4,1), (4,2), (4,3), (4,4); number $= 7$ (1); $P = \frac{7}{36}$ (1).	1+1
13	Total possible outcomes $= 22$; Favourable numbers: 3, 5, 6, 9, 10, 12, 15, 18, 20, 21; number $= 10$ (1); $P = \frac{10}{22} = \frac{5}{11}$ (1).	1+1
14	Total balls $= 3 + 7 + 5 = 15$; Favourable balls $= 3$ (1); $P = \frac{3}{15} = \frac{1}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{blue}) = \frac{x}{5+x}$ and $P(\text{red}) = \frac{5}{5+x}$ (1); $\frac{x}{5+x} = 3 \times \frac{5}{5+x}$ gives $x = 3 \times 5$ (1); $x = 15$ (1).	1+1+1
16	$\frac{6+x}{16+x} = \frac{4}{9}$ (1); Cross-multiplying: $9(6 + x) = 4(16 + x)$ (1); $x = 2$ (1).	1+1+1
17	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (4,6), (5,5), (5,6), (6,4), (6,5), (6,6); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1+1
	OR: $P(\text{blue}) = \frac{x}{12+x}$ and $P(\text{red}) = \frac{12}{12+x}$ (1); $\frac{x}{12+x} = 2 \times \frac{12}{12+x}$ gives $x = 2 \times 12$ (1); $x = 24$ (1).	1+1+1
18	$\frac{8+x}{12+x} = \frac{3}{4}$ (1); Cross-multiplying: $4(8 + x) = 3(12 + x)$ (1); $x = 4$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 6; 4; 4 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{9}$, $\frac{1}{9}$ (1).	2+2+1
	OR: Total outcomes $= 45$ (2); Favourable counts: 9, 14, 11 (2); Probabilities: $\frac{1}{5}$, $\frac{14}{45}$, $\frac{11}{45}$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 9; 8; 9 (2); Probabilities: $\frac{1}{4}$, $\frac{2}{9}$, $\frac{1}{4}$ (1).	2+2+1
	OR: Total outcomes $= 30$ (2); Favourable counts: 14, 5, 9 (2); Probabilities: $\frac{7}{15}$, $\frac{1}{6}$, $\frac{3}{10}$ (1).	2+2+1

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